

make other mechanisms such as reputation, unreliable compared to staked balances.⁷

SWAP

A swap is simply the exchange of one type of token for another. The key benefit of swapping in DeFi is that it is atomic and non-custodial. Funds can be custodied in a smart contract with withdrawal rights that can be exercised anytime before the swap is completed. The swap executes only when the exchange conditions are agreed to and met by all parties and are enforced by the smart contract. If any condition is not met, the entire transaction is canceled and all parties retain their custodied funds. A platform that facilitates token swapping on Ethereum in a non-custodial fashion is a *decentralized exchange* (DEX). There are two primary mechanisms for DEX liquidity: an order-matching approach and an *Automated Market Maker* (AMM).

Order-Book Matching

Order-book matching is a system in which all parties must agree on the swap exchange rate. Market makers can post bids and asks to a DEX, and allow takers to fill the quotes at the previously agreed on price. Until the offer is taken, the market maker retains the right to remove the offer or update the exchange rate as market conditions change.

The order-matching approach is expensive and inefficient because each update requires an on-chain transaction. An